

Professional Vue.js

Version 2.0, September 2025

In this 2-day course, students will start with an optional review of advanced JavaScript, followed by the fundamental concepts and structures of Vue.js development. Students will use Vue.js to build a real-world frontend application through a series of hands-on labs.

TARGET AUDIENCE

This course is designed for experienced front-end web developers who wish to begin using Vue.js.

OBJECTIVES

After taking this course, students will be able to:

- Install, configure, and use modern web tooling
- Learn to create test suites for Vue
- Understand what Vue.js is and what problem it solves
- Explore the basic architecture of a Vue.js application
- Gain a deep knowledge of Vue.js components
- Build a Single Page Application with Vue Router
- Use Vuex for maintaining state in a Vue.js application
- Learn Vue.js best practices

TOPICS

- Modern JavaScript tooling
- Debugging JavaScript
- Object-oriented JavaScript
- All the basic Vue capabilities and features
- Communication between components
- Data binding
- Communication with the server via Ajax
- State management with Vuex
- Creating SPAs with Vue
- Asynchronous JavaScript - promises, async/await (time permitting)
- Animation and Transitions in Vue (time permitting)

Prerequisites

Before taking this course, students should have solid experience with HTML5, CSS3, and JavaScript

Duration

2 days

OUTLINE

1. JavaScript Review (Optional)
 - a. JavaScript Syntax
 - b. JavaScript Data Types
 - c. Variable Scoping
 - d. JavaScript Operators

- e. Template Literals
 - f. Functions
 - g. Arrow Functions
 - h. JavaScript Objects
 - i. Prototypal Inheritance
 - j. Default Parameter Handling
 - k. Rest and Spread
 - l. Tagged Template Literals
 - m. Enhanced Object Properties
 - n. Method Notation
 - o. Array Matching
 - p. Object Matching
 - q. Symbol Primitive
 - r. User-defined Iterators
 - s. For-Of Operator
 - t. Creating and Consuming Generator Functions
 - u. Class Syntax
 - v. Understanding this
 - w. Array Methods
 - x. Promises
 - y. Async / Await
2. Vue.js
- a. Virtual DOM
 - b. Lab 01: Get Started with Vite
 - c. Lab 02: Your First Component
 - d. Lab 03: Create More Components
 - e. Lab 04: Testing Vue
 - f. Lab 05: Manual In-Browser Testing and Debugging
 - g. The Vue Instance
 - h. Instance Properties and Methods
 - i. Vue Template Syntax
 - j. Vue.js Directives
 - k. Loops and Lists
 - l. Lab 06: Static Version
 - m. Using Filters
 - n. Adding Styles Conditionally
 - o. Binding Styles
 - p. Outputting Raw HTML
 - q. Lab 07: Styling Vue Components
 - r. Computed Properties
 - s. Lab 08: Computed Properties
 - t. Vue State
 - u. Lab 09: Methods and State
 - v. Event Handling

- w. Lab 10: Events
- x. Watchers
- y. Vue Instance Lifecycle
- z. Using Lifecycle Hooks
- aa. Lab 11: Component Lifecycle
- bb. Vue and Forms
- cc. Lab 12: Forms
- dd. Vue Components
- ee. Content Distribution with Slots
- ff. Lab 13: Slots
- gg. Using Vuex
- hh. Lab 14: Implementing Vuex
- ii. Creating an SPA with Vue-Router
- jj. Lab 15: Routing
- kk. Lab 16: AJAX with Vuex
- ll. Transitions and Animations
- mm. Optional Lab: Transitions and Animation